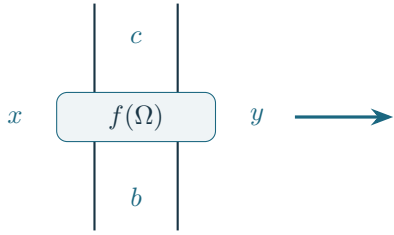
One homogeneous polynomial for every original coupon



Original input and output faces
with their physical head conventions

$$H_f(x,b,c,y) \;=\; \sum_{i+k+l+t=d} a_f(i,k,l,t)\, b^i x^k y^l c^t$$

$$d \;=\; N-1, \quad i,k,l,t \;\geq\; 0$$

The coefficients retain the actual phases
and repeated Casimir-node data.

Multiply these polynomials in the original connected fixed flat plat.
Every shared face uses one common variable.

$$Z_N \;=\; \left[\prod_{f \text{ ordinary}} L_f^d\right] \prod_e H_{f_e}\Big|_{L_u=L_v=0}$$

The original mark gives the two zero targets u, v.

Total degree dR is already consumed by the R ordinary targets.
Every other Fourier alias is excluded, including disconnected replacements.

Exact at roots of unity and arbitrary fixed bridge width.
A volume limit still requires the stated geometric and source estimates.